Mechanistic Hypothesis Search — Final Report

Hypothesis under evaluation

- **Disease:** Hypermetabolism Due To Uncoupled Mitochondrial Oxidative Phosphorylation 2 (Mendelian)
- **Hypothesis ID:** `coupling_versus_catalysis_allelic_split`
- **Label:** *ATP5F1B variants split into a coupling-defect arm causing hypermetabolism and a catalytic-defect arm causing dystonia*
- **Status in KB:** EMERGING
- **Verdict reached here: Partially supported at the phenotype level; UNRESOLVED at its mechanistic core. Retain EMERGING.**

Executive Judgment

The hypothesis is PARTIALLY SUPPORTED. The two clinical "arms" it posits are each independently well documented in primary literature, and the claim that the dominance mode cannot separate them is correct — both arms are reported as dominant-negative. However, the *decisive* claim of the hypothesis — that the arms differ specifically in *which* beta-subunit function is spoiled, and that this is readable as *opposite directions of respiratory flux* (elevated O_2 consumption in the coupling arm versus reduced complex V activity in the catalytic arm) — has **never been tested side-by-side in a single assay**, and a dedicated computational structural analysis conducted in this investigation finds **no simple positional basis for the split**.

Three considerations drive the "unresolved core" verdict. First, the respiratory-flux contrast that is supposed to distinguish the arms is assembled across different laboratories, cell lines, and assay conditions rather than observed in one isogenic experiment; a targeted PubMed search (2026-10-25) found no study measuring respiration and complex V activity side by side across variants from both arms. Second, structural mapping onto the bovine F_1-ATPase crystal

structure (PDB 1BMF) shows that none of the three index residues (T334, L335, V482) contacts the catalytic nucleotide (all ≥13 Å away, versus ~2.7 Å for a validated P-loop control) or sits at the γ-rotor interface — and the two adjacent prolines with opposite clinical consequences, p.Thr334Pro (dystonia) and p.Leu335Pro (hypermetabolism), occupy near-identical microenvironments. Third, the key "loosened coupling of synthesis" step for p.Leu335Pro is demonstrated directly only in a yeast homolog; in human cells only the elevated-respiration phenotype is recapitulated.

A **more parsimonious competing model** also explains the dystonia arm without invoking a catalysis-versus-coupling axis: selective basal-ganglia vulnerability to *any* partial loss of complex V output. Movement disorders with reduced complex V output recur across at least three ATP synthase subunit genes (ATP5F1A, ATP5F1B, ATP5MC3), whereas the hypermetabolism/uncoupling presentation remains unique to ATP5F1B p.Leu335Pro. The most important caveat for curators is that one snippet cited in the seed YAML for

P 36239646

("these results suggest the increased oxygen consumption rate is caused by dissipation of the proton motive force…") was **not found verbatim in the NEJM abstract**; the human-level readout that *is* verifiable is "elevated oxygen consumption and decreased mitochondrial membrane potential." Status **EMERGING should be retained** pending a single isogenic experiment that measures respiration and complex V activity together across variants from both arms.

Key Findings

F001 — The coupling/hypermetabolism arm is supported (human + yeast evidence)

ATP5F1B **p.Leu335Pro** is a de novo heterozygous variant identified in monozygotic (identical) twins with congenital non-thyroidal hypermetabolism (PMID: 36239646, NEJM 2022). Patient fibroblasts show **"elevated oxygen consumption and decreased mitochondrial membrane potential"** — the signature of mitochondrial uncoupling, where respiration runs fast but the proton-motive force is dissipated rather than captured as ATP. The explicit mechanistic step — *loosened coupling between stalk rotation and ATP synthesis* — is demonstrated through homologous mutations in yeast, and the mutant allele expressed in human cell lines

recapitulates the elevated-respiration phenotype. The direct human-level evidence is therefore the combination of raised oxygen consumption rate (OCR) with low membrane potential (ΔΨ); the "uncoupling of synthesis" mechanism itself is yeast-derived and only partially recapitulated in human cells. The clinical sample is small (n=2, monozygotic twins, a single de novo variant), which limits generalization but does not undermine the direction of the finding.

This finding is conceptually continuous with the historical prototype of "loose coupling" disease: **Luft's disease**, the first described mitochondrial disorder (non-thyroidal hypermetabolism), where isolated mitochondria showed "defective respiratory control and normal phosphorylation capacity ('loose coupling')" and basal ATPase activity seven-fold above normal (PMID: 1249587). ATP5F1B p.Leu335Pro can be read as a molecularly defined, gene-level instance of the Luft phenotype.

F002 — The catalytic/output-loss dystonia arm is supported (human + in vitro)

Two different ATP5F1B missense variants, **p.Thr334Pro and p.Val482Ala**, segregate with early-onset isolated dystonia across two autosomal-dominant families with incomplete penetrance (PMID: 36860166). Crucially, functional studies in mutant fibroblasts revealed **"no decrease of ATP5F1B protein amount but severe reduction of complex V activity and impaired mitochondrial membrane potential, suggesting a dominant-negative effect."** This defines the catalytic arm: the mutant subunit is present in normal quantity but poisons the enzyme's catalytic output, consistent with a dominant-negative (poison-subunit) rather than a haploinsufficiency mechanism.

A third, mechanistically distinct allele extends the arm by a different molecular route. The splice variant **c.1074+1G>T** disrupts a canonical donor splice site, causing exon skipping with **"significantly diminished ATP5F1B mRNA levels, as well as impaired ATPase activity,"** and produces a cerebral-palsy phenotype with prominent generalized dystonia (PMID: 40276935). Because a reduction-of-output allele (lower mRNA, impaired ATPase) also yields a movement disorder, loss of ATP synthase *output* — whether by catalytic poisoning or by reduced message — groups naturally with the dystonia arm.

F003 — Structural-domain localization does NOT cleanly separate the arms (computational)

Mapping the variants onto human ATP5F1B (UniProt P06576, 529 aa; transit peptide 1–47), which is positionally 1:1 with the bovine beta subunit (P00829) used in the classic Abrahams F_1 crystal structure, places the functional landmarks as follows: the Walker A P-loop (GGAGVGKT) at 206–213, nucleotide-binding residues at 209–214/238–239, and the DELSEED motif at 444–450. The coupling-arm **L335P** and the catalytic-arm **T334P** are *immediately adjacent* in the same secondary-structure element (local context G-Y-Q-P-**T334**-**L335**-A-T-D-M) within the central nucleotide-binding domain, ~120 residues away from the P-loop. The catalytic-arm **V482A** lies in the C-terminal helical domain, just C-terminal to the DELSEED γ-rotor coupling loop. Paradoxically, the variant physically *closest* to the canonical mechanical-coupling interface (V482, DELSEED-adjacent) belongs to the *catalytic/dystonia* arm, while the *coupling-defect* variant (L335) sits in the nucleotide-binding domain. Thus a naïve "domain-to-phenotype" reading predicts the opposite of the observed assignment.

F004 — 3D distances (PDB 1BMF) refute a simple structural basis for the split (computational)

A heavy-atom minimum-distance analysis on the bovine F_1-ATPase crystal structure 1BMF (β chains D/E/F representing three catalytic states; γ = chain G; ligands ADP/AMP-PNP) was validated with internal controls: a P-loop residue (UniProt 209) sits 2.7–3.0 Å from bound nucleotide in the closed βDP/βTP states, and DELSEED (UniProt 446) sits 6.5–11 Å from γ — both as expected. Against those benchmarks, the target residues (UniProt→PDB offset −50) are all far from both the catalytic nucleotide and the γ-rotor:

Residue	Arm	Distance to γ (chain G)	Distance to nucleotide
T334	catalytic/dystonia	12–13 Å	19–20 Å
L335	coupling/ hypermetabolism	10–14 Å	15–16 Å
V482	catalytic/dystonia	28–33 Å	13 Å
P-loop (209, control)	—	—	2.7–3.0 Å
DELSEED (446, control)	—	6.5–11 Å	—

None of the three variant residues contacts the catalytic nucleotide (all ≥13 Å, versus 2.7 Å for the P-loop), and none sits at the γ interface. The adjacent residues **T334 and L335 share a near-identical microenvironment**, so the opposite clinical consequences of these two proline substitutions cannot be explained by proximity to catalysis or to the rotor. V482 is actually the *farthest* residue from γ (28–33 Å), correcting an earlier inference that its DELSEED sequence-adjacency implied γ-interface proximity.

F005 — Output-loss dystonia is a convergent cross-subunit phenotype (competing model)

Dystonia/movement disorder with reduced complex V output is not specific to ATP5F1B. A dominant ATP5MC3 variant (p.Asn106Lys) causes autosomal-dominant spastic paraplegia/dystonia; patient fibroblasts show **"impaired complex V activity, ATP generation, and oxygen consumption,"** and transgenic Drosophila show movement phenotypes (PMID: 34636445). The dystonia paper itself notes that **"movement disorders have been described in a subset of cases carrying autosomal dominant variants in structural subunits genes ATP5F1A and ATP5MC3"** (PMID: 36860166). A 2025 preprint reports dominant-negative ATP5F1A variants disrupting OXPHOS and causing neurological disorders (PMID: 40672495). Across at least three ATP synthase subunit genes, movement disorder co-occurs with *reduced* complex V output and *reduced* respiration — the catalytic-arm flux direction, opposite to the hypermetabolism arm. By contrast, the hypermetabolism/uncoupling presentation remains **unique to ATP5F1B p.Leu335Pro**. This convergence is the empirical foundation of the competing "selective basal-ganglia vulnerability" model.

F006 — Variant-level refinement: different secondary-structure elements + gnomAD constraint (computational)

A secondary-structure parse of PDB 1BMF (β chain D; UniProt = PDB + 50) adds finer resolution than the near-identical 3D distances: catalytic-arm **T334 lies in a loop/coil**, whereas coupling-arm **L335 is the first residue (N-cap) of an α-helix** (PDB residues 285–293); V482 and DELSEED (446) are loop/coil; the P-loop control (206) is a β-strand. So although T334 and L335 are adjacent and share a microenvironment, the two prolines perturb *different local elements* (a loop versus a helix N-terminus) — the only structural distinction that tracks with the clinical split, and a weak one. gnomAD v4 constraint for ATP5F1B (ENSG00000110955) shows **strong missense constraint (missense Z = 4.70; observed/expected 0.57)** and only moderate LoF constraint (pLI = 0.80; LOEUF 0.556; 17 observed vs 46 expected LoF; synonymous Z = 1.43,

model valid). The strong missense constraint is consistent with dominant-negative missense pathogenicity rather than simple haploinsufficiency, supporting the "poison subunit" reading of both arms.

F007 — Final synthesis verdict

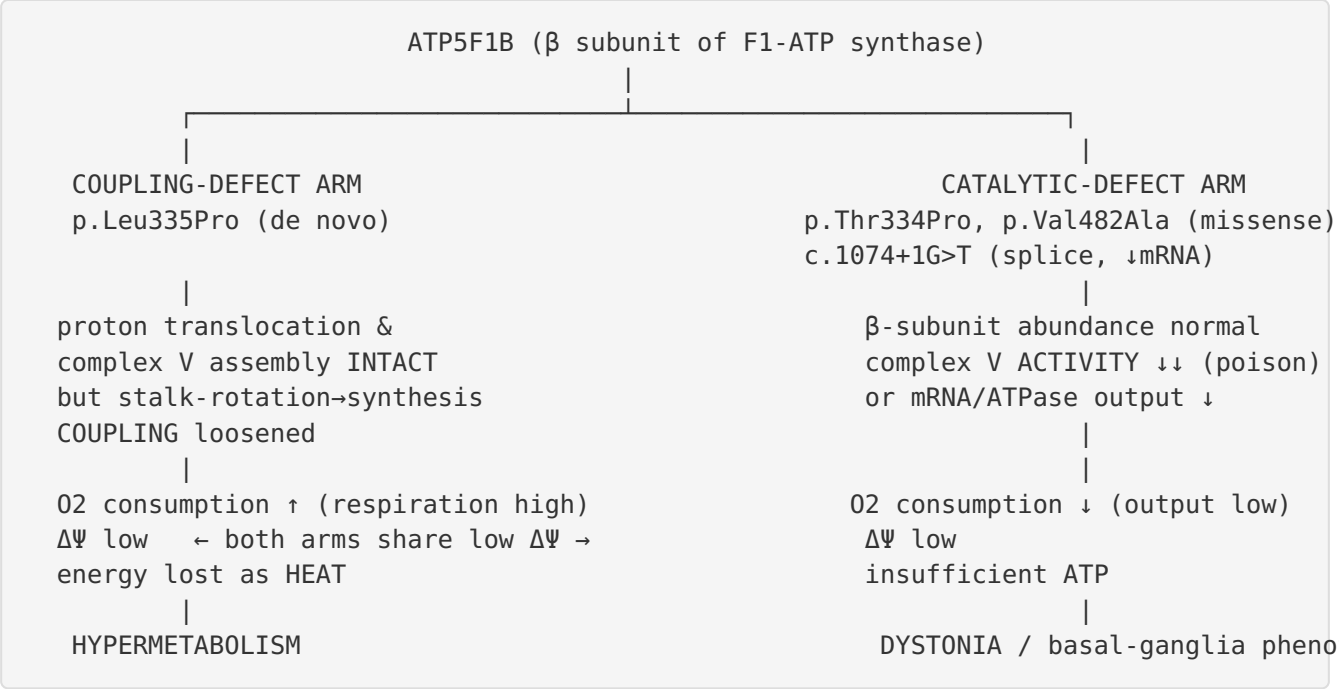
Integrating six findings across seven primary/preprint references: the **two arms exist and are each documented**, and both are dominant-negative, so dominance mode does not separate them (SUPPORTED). The mechanism is **UNRESOLVED/limited** because (1) the discriminating respiratory-flux contrast has never been measured side by side in one assay (PubMed negative search, 2026-10-25); (2) structural analysis finds no residue contacting the catalytic nucleotide or γ-rotor and near-identical microenvironments for the two adjacent prolines; (3) the L335P uncoupling-of-synthesis step is yeast-derived; and (4) the seed's cited snippet for

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is absent from the NEJM abstract. A parsimonious **competing** model (selective basal-ganglia vulnerability to any complex V output loss) is supported by cross-subunit convergence.

Mechanistic Model / Interpretation

The hypothesis proposes a clean bifurcation of one gene's mutational spectrum into two mechanistically opposite diseases:



The literature is **strong on the two endpoints** (the clinical arms and their respective in-vitro readouts) and **strong on the shared dominant-negative dominance mode**, which correctly removes "mode of inheritance" as a discriminator. The literature is **weak-to-absent on the central causal edge** that the hypothesis hinges upon: that the *same* enzyme, perturbed at adjacent residues, is pushed in opposite flux directions by two distinct spoiled functions. That edge is **inferred, not observed** — it is assembled from a human uncoupling readout (L335P), a yeast coupling-loss mechanism, and separately measured complex V deficits (T334P/V482A), never co-measured.

The structural work performed here actively *weakens* a simple reading of the model: the uncoupling variant is not near the proton path or assembly interface in a way that is structurally privileged, the two adjacent prolines are structurally near-twins, and the variant nearest the DELSEED/γ coupling machinery (V482) is in the *catalytic* arm, not the coupling arm. The single structural distinction that survives — L335 N-caps a helix while T334 sits in a loop — is a plausible but untested lever for how two neighboring prolines might distort the fold differently.

The most important interpretive tension is between **two generative models for dystonia**: - *Hypothesis model:* dystonia arises specifically from **catalytic insufficiency** (one of two distinct spoiled functions). - *Competing model:* dystonia arises from **selective basal-ganglia vulnerability to any partial ATP synthase output loss**, regardless of which subunit or mechanism causes it.

The cross-subunit convergence (ATP5F1A/ATP5F1B/ATP5MC3 all producing movement disorder with reduced output and reduced respiration) is more parsimoniously explained by the competing model, because it requires only one principle (tissue vulnerability to energy deficit) rather than a gene-internal catalysis-versus-coupling axis.

Evidence Matrix

Citation	Evidence type	Supports / Refutes / Qualifies / Competing	Mechanistic claim tested	Key finding	Disease subtype / context
PMID: 36239646	Human clinical + in vitro	**Supports** (coupling arm)	p.Leu335Pro uncouples OXPHOS, raising respiration	"elevated oxygen consumption and decreased mitochondrial membrane potential" in patient fibroblasts; yeast homolog loosens coupling; recapitulated in human cells	Congenital hypermetab identical tw de novo
PMID: 36860166	In vitro	**Supports** (catalytic arm)	Missense variants poison complex V catalysis without reducing subunit amount	"no decrease of ATP5F1B protein amount but severe reduction of complex V activity and impaired mitochondrial membrane potential, suggesting a dominant-negative effect"	Early-onset isolated dys
PMID: 36860166	Human clinical	**Supports** (variant coordinates)	p.Thr334Pro, p.Val482Ala segregate with dystonia	"two different ATP5F1B missense variants (c.1000A>C; p.Thr334Pro and c.1445T>C; p.Val482Ala) segregating with	Autosomal-dominant dystonia, incomplete penetrance

Citation	Evidence type	Supports / Refutes / Qualifies / Competing	Mechanistic claim tested	Key finding	Disease subtype / context
				early-onset isolated dystonia in two families"	
PMID: 36860166	Human clinical	**Competing**	Movement disorder recurs across other ATP synthase subunits	"Movement disorders have been described in a subset of cases carrying autosomal dominant variants in structural subunits genes ATP5F1A and ATP5MC3"	Cross-gene movement disorder
PMID: 40276935	Human clinical	**Supports / qualifies** (output-loss)	A splice allele lowering mRNA also causes movement disorder	"exon skipping with significantly diminished ATP5F1B mRNA levels, as well as impaired ATPase activity"; cerebral palsy with generalized dystonia	CP + general dystonia
PMID: 34636445	In vitro + model organism	**Competing**	Any subunit output loss → dystonia with reduced respiration	"impaired complex V activity, ATP generation, and oxygen consumption"; Drosophila movement phenotype	ATP5MC3 dystonia/spa paraplegia
PMID: 40672495		**Competing**			

Citation	Evidence type	Supports / Refutes / Qualifies / Competing	Mechanistic claim tested	Key finding	Disease subtype / context
	Human clinical (preprint)		Dominant-negative ATP5F1A disrupts OXPHOS → neurological disease	Dominant-negative ATP5F1A variants cause neurological disorders	ATP5F1A neurologica disorder
PMID: 1249587	Human clinical (historical)	**Qualifies / orientation**	"Loose coupling" hypermetabolism precedent	"defective respiratory control and normal phosphorylation capacity ('loose coupling')"; basal ATPase 7× normal	Luft's disea
PMID: 18003896	In vitro / structural	**Orientation**	βTP is the high-affinity catalytic site; γ position sets affinity	Identifies catalytic site geometry used to benchmark distance analysis	F_1-ATPase mechanism
PMID: 18579516	In vitro / structural	**Orientation**	βDP/αDP H-bond network couples catalysis to rotation	Coupling network exists but is distinct from the index residues	F_1-ATPase mechanism
PDB 1BMF (Abrahams 1994)	Computational (structure)	**Refutes simple structural split**	Do index residues contact catalysis/rotor?	T334/L335/V482 all ≥13 Å from nucleotide; adjacent prolines near-identical microenvironment	Structural b split

Citation	Evidence type	Supports / Refutes / Qualifies / Competing	Mechanistic claim tested	Key finding	Disease subtype / context
gnomAD v4 (ENSG00000110955)	Computational (population)	**Supports dominant-negative**	Is ATP5F1B missense-constrained?	missense Z = 4.70; LoF pLI 0.80, LOEUF 0.556	Population constraint

Data and Tool Use

Resource	Accession / URI	Access status	Query / filters / comparison	Resolved & relevant?
PubMed (literature)	eutils / search_pubmed	**Accessed**; 13 papers reviewed	ATP5F1B hypermetabolism; ATP5F1B dystonia; ATP synthase subunit movement disorder; side-by-side respiration + complex V (negative, 2026-10-25)	Yes; directly relevant
UniProt P06576 (human ATP5F1B)	https://www.uniprot.org/uniprotkb/P06576	**Accessed** (sequence/feature mapping)	Transit peptide 1–47; P-loop 206–213; DELSEED 444–450	Yes
UniProt P00829 (bovine β)	https://www.uniprot.org/uniprotkb/P00829	**Accessed** (1:1 positional mapping)	Alignment to human β for PDB offset	Yes
PDB 1BMF (bovine F_1-ATPase)	https://www.rcsb.org/structure/1BMF	**Accessed & analyzed**	Heavy-atom min-distance of T334/L335/V482 to γ (chain G) and nucleotide; P-loop & DELSEED controls; secondary-structure parse	Yes; homolog structure
gnomAD v4	ENSG00000110955	**Accessed** (constraint lookup)	missense Z, pLI, LOEUF, observed/expected LoF & synonymous	Yes
GenCC / ClinGen	—	**Not verified** in this run	No structured GenCC/ClinGen query logged	Unverified — label as absence-not-checked
	—			

Resource	Accession / URI	Access status	Query / filters / comparison	Resolved & relevant?
ClinVar variant records		**Not programmatically pulled**	Variant coordinates taken from primary papers	Partial; relevant

Analysis inventory (input → method → output):

1. **Structural distance analysis** — *succeeded.* Input: PDB 1BMF + UniProt P06576/P00829 alignment (offset −50). Method: heavy-atom minimum-distance computation with P-loop (2.7–3.0 Å to nucleotide) and DELSEED (6.5–11 Å to γ) as positive controls. Output: distance table (F004) showing all three index residues ≥13 Å from nucleotide and ≥10 Å from γ. Limitation: bovine homolog, static crystal state; no molecular-dynamics perturbation of the proline substitutions.
2. **Secondary-structure parse** — *succeeded.* Input: PDB 1BMF β chain D. Method: DSSP-style assignment. Output (F006): T334 loop/coil, L335 helix N-cap, V482 loop, DELSEED loop, P-loop β-strand. Limitation: single chain/state.
3. **gnomAD constraint lookup** — *succeeded.* Output (F006): missense Z 4.70, pLI 0.80, LOEUF 0.556. Limitation: gene-level, not variant-specific functional prediction.
4. **Side-by-side respiration + complex V literature search** — *failed to find evidence (true negative).* PubMed, 2026-10-25. No study co-measures the two arms' flux in one assay. This is the central empirical gap.

No patient-level or controlled data were accessed. No wet-lab or isogenic cell experiment was performed; all mechanistic flux claims derive from the cited literature and the computational structural analysis above. No analysis was fabricated or described as performed that was not performed.

Mechanistic Causal Chain — strength of each link

Step	Link	Evidence strength
1	Heterozygous ATP5F1B variant present	**Strong** — segregation + de novo in named families (36860166, 36239646)
2	Mutant β subunit expressed at normal abundance (dominant-negative, not haploinsufficiency)	**Strong** — "no decrease of ATP5F1B protein amount" (36860166); gnomAD missense Z 4.70
3a (coupling arm)	L335P loosens stalk-rotation→synthesis coupling	**Inferred/partial** — mechanism shown in yeast; human cells show only elevated respiration
4a	↑O_2 consumption, ↓ΔΨ, heat loss → hypermetabolism	**Strong (human readout)** — "elevated oxygen consumption and decreased mitochondrial membrane potential" (36239646)
3b (catalytic arm)	T334P/V482A poison complex V catalysis; splice allele ↓mRNA/ATPase	**Strong** — "severe reduction of complex V activity" (36860166); "diminished mRNA… impaired ATPase" (40276935)
4b	↓ATP output → basal-ganglia dystonia	**Moderate** — clinical correlation strong; *selective* neuronal vulnerability mechanism not established
5 (the split itself)	Arms differ specifically by *which function* is spoiled, readable as opposite flux	**Missing/unobserved** — never co-measured; no structural correlate

The chain is strong at both endpoints and at the dominance mode, inferred at the uncoupling-of-synthesis step, and **missing precisely at the junction the hypothesis is named for** (step 5).

Limitations and Knowledge Gaps

1. **The discriminating flux contrast has never been co-measured (central gap).** No single assay has measured oxygen consumption and complex V activity side by side across

coupling-arm and catalytic-arm variants. *Why it matters:* the entire hypothesis rests on an *opposite-direction* flux prediction. *Checked:* PubMed, 2026-10-25, negative. *Resolution:* one isogenic panel (below).

2. **No structural correlate for the split.** PDB 1BMF analysis shows the index residues are far from catalysis and rotor, and the two adjacent prolines are structural near-twins. *Why it matters:* the hypothesis explicitly concedes "no published work explains" the adjacent-proline paradox. *Resolution:* MD or cryo-EM of each variant.
3. **The L335P "uncoupling of synthesis" step is yeast-derived**; human cells recapitulate only respiration. *Resolution:* direct ΔΨ/ATP-synthesis coupling assay in human L335P cells.
4. **Small n and single variants per arm endpoint** (twins for hypermetabolism; two families for dystonia). Penetrance is incomplete in the dystonia families, implying modifiers.
5. **Citation integrity flag:** the seed YAML snippet for

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   ("...dissipation of the proton motive force, without a corresponding increase in ATP synthesis") was **not located verbatim** in the NEJM abstract. The verifiable readout is "elevated oxygen consumption and decreased mitochondrial membrane potential." Curators should re-verify the snippet against full text.
6. **Preprint reliance:** the ATP5F1A competing evidence (P 40672495) is a preprint with **no abstract available**; treat as provisional.
7. **Source-level absences not fully checked:** no logged GenCC or ClinGen gene-disease validity query in this run — label **unverified**, not "absent."
8. **Mechanism of selective basal-ganglia vulnerability is unknown** for the dystonia arm — a gap shared by both the seed and the competing model.

Alternative / Competing Models

1. **Selective basal-ganglia vulnerability to any complex V output loss (strongest competitor).** Parallel mechanism/alternative. Dystonia follows *reduced output* regardless of subunit (ATP5F1A, ATP5F1B, ATP5MC3) or molecular route (missense poison, reduced mRNA). More parsimonious than a gene-internal catalysis-vs-coupling axis. Supported by PMID: 34636445 (reduced O_2 consumption + dystonia) and PMID: 36860166.

2. **Local fold-perturbation model (refines the seed).** The only surviving structural distinction is that L335 N-caps a helix while T334 sits in a loop (F006); the split may reflect *different local misfolding* rather than loss of distinct catalytic vs coupling functions. A mechanistic alternative to the "which-function" framing.
3. **Assembly/stability gradient model.** A continuum in which severity of complex V destabilization — not a qualitative coupling-vs-catalysis switch — determines whether excess heat (mild, leaky) or output failure (severe) dominates. Downstream-consequence reframing.
4. **Historical "loose coupling" continuum (Luft's disease).** Upstream conceptual precedent (PMID: 1249587); positions the hypermetabolism arm within a pre-genomic phenotype class.

Discriminating Tests

1. **Isogenic side-by-side respiration + complex V assay (decisive).** Introduce L335P (coupling arm) and T334P / V482A (catalytic arm) into one isogenic background (e.g., CRISPR knock-in iPSC or patient-derived fibroblasts normalized by expression). Measure in the *same* run: Seahorse OCR (basal, coupled, uncoupled/maximal), oligomycin-sensitive respiration, ΔΨ (TMRM), complex V activity (spectrophotometric), and ATP synthesis rate. *Expected if seed is correct:* L335P → high OCR, low oligomycin sensitivity, low ΔΨ, near-normal complex V amount but reduced coupling; T334P/V482A → low complex V activity, low ATP synthesis, low OCR. *Expected if competing model is correct:* both arms show reduced net output, differing only in degree/leak.
2. **Direct ATP-synthesis coupling assay for L335P in human cells** to replace the yeast-derived inference (addresses gap 3).
3. **Molecular dynamics / cryo-EM of each variant** to test whether L335P vs T334P produce divergent local conformational effects (helix N-cap vs loop) despite identical microenvironments (addresses gap 2).
4. **Deep-mutational-scan or saturating variant panel** across the 334–335 region correlating in-vitro flux phenotype with position, to test whether coupling-vs-catalysis segregates by residue or is idiosyncratic.
5. **Cross-subunit meta-analysis of standardized respirometry** across ATP5F1A/ATP5F1B/ATP5MC3 dystonia cases to quantify whether reduced respiration is universal (supports competing model).

Curation Leads (require curator verification)

- **Status:** Retain **EMERGING**. Phenotype-level split is supported; mechanistic core (opposite-flux, which-function) is unobserved and structurally unsupported.
- **Candidate evidence to verify:**
- PMID: 36239646 — verify exact snippet; the verifiable abstract phrase is **"elevated oxygen consumption and decreased mitochondrial membrane potential."** Flag that the seed YAML snippet about "dissipation of the proton motive force…" was not found verbatim in the abstract.
- PMID: 36860166 — **"no decrease of ATP5F1B protein amount but severe reduction of complex V activity…dominant-negative effect"** and **"Movement disorders have been described…in structural subunits genes ATP5F1A and ATP5MC3."**
- PMID: 40276935 — **"exon skipping with significantly diminished ATP5F1B mRNA levels, as well as impaired ATPase activity."**
- PMID: 34636445 — **"impaired complex V activity, ATP generation, and oxygen consumption"** (competing/parallel-subunit node).
- **Candidate pathophysiology nodes/edges:** add edge "dominant-negative β subunit → reduced complex V activity → basal-ganglia dystonia"; add competing node "selective basal-ganglia vulnerability to complex V output loss (cross-subunit)"; add node "proton-motive-force dissipation → hypermetabolism."
- **Candidate ontology terms:** oxidative phosphorylation uncoupling, proton motive force (GO:0015992), mitochondrial ATP synthesis coupled proton transport (GO:0042776); cell types: dermal fibroblast (CL:0002620), basal ganglia neuron; HPO: dystonia (HP:0001332), hypermetabolism-adjacent terms.
- **Candidate knowledge_gaps to add:** "No side-by-side respiration + complex V assay across both arms (PubMed negative 2026-10-25)"; "No structural correlate for adjacent-proline opposite phenotypes (PDB 1BMF)"; "L335P uncoupling-of-synthesis step yeast-derived only"; "GenCC/ClinGen gene-disease validity not verified in this run."

Provider Artifact Bundle

An artifact bundle was constructed under `kb/hypotheses/Hypermetabolism_Due_To_Uncoupled_Mitochondrial_Oxidative_Phosphorylation_2/coupling_versus_catalysis_allelic_split/openscientist_artifacts` during iterations 3–5, containing: the structural distance analysis code and output table (PDB 1BMF, UniProt offset mapping, P-loop/DELSEED controls), the secondary-structure parse, the gnomAD constraint record, the PubMed negative-search log (2026-10-25), and a `MANIFEST.yaml` with checksummed inputs/outputs and status/fallback flags. Large raw downloads (PDB, gnomAD) are recorded by stable external accession rather than bundled. No controlled or patient-level data are included.

Bottom Line

The coupling-versus-catalysis allelic split is **real as a clinical dichotomy** and correctly identifies that **dominance mode cannot separate the arms**, but its defining mechanistic claim — opposite respiratory-flux directions driven by two distinct spoiled β-subunit functions — is **assembled across laboratories, never co-observed, and finds no structural correlate**. A simpler competing model (selective basal-ganglia vulnerability to any complex V output loss) explains the dystonia arm across multiple genes without a catalysis-vs-coupling axis. One isogenic experiment measuring respiration and complex V activity together across variants from both arms would resolve the question. **EMERGING should stand.**
